

MUHAMMAD AFNAN SIDDIQUI

GitHub: github.com/afnan-nex

Portfolio: afnan-nex.github.io/portfolio/index.html

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PROFESSIONAL SUMMARY

Full-stack developer with expertise in cross-platform application development, desktop and mobile software engineering, and automation workflows. Experienced in building inventory management systems, payment processing applications, and developer tooling. Proficient in multiple programming languages and frameworks with demonstrated ability to design scalable database architectures and implement role-based access control systems.

TECHNICAL SKILLS

Programming Languages

C++, Python, JavaScript, Dart, SQL

Web Development

HTML5, CSS3, Node.js, REST API Integration, Express.js

Mobile Development

Flutter, Dart, Cross-platform Application Development, Android Development

Desktop Development

Qt6, CustomTkinter, PyInstaller, C++, GUI Development

Database Technologies

SQLite, PostgreSQL, SQL, Database Design, Database Normalization, Supabase

Backend and APIs

Node.js, Express, Authentication, Authorization, Role-Based Access Control (RBAC), CRUD Applications

Core Competencies

Offline-first Application Design, POS System Architecture, Inventory Management, File Handling, Barcode Scanner Integration, Receipt Printing Workflows, Local Network Communication, API Integration, Command-Line Automation, Cross-platform Software Development, Debugging and Troubleshooting, UI/UX Implementation, Software Deployment Planning

Version Control and Collaboration

Git, GitHub, Feature-first Project Organization, Git-based Project Management

Developer Tools and Environments

Visual Studio Code, Qt Creator, Android Studio, Flutter SDK, Node.js, npm, Python, CMake, Ninja, Windows Terminal, PowerShell, Command Prompt, ADB Platform Tools

Automation and DevOps

Package Management Automation, PowerShell Scripting, Windows Batch Scripting, FFmpeg, yt-dlp, aria2, Custom Script Development

Web Infrastructure

WordPress, WooCommerce, Netlify, Google Cloud Platform (GCP), Hugging Face, n8n Automation Platform

AI and Machine Learning Integration

OpenRouter, Hugging Face Models, Pollinations AI, Google Gemma, Qwen, DeepSeek AI, Leonardo AI, Unreal Speech API, Kokoro TTS, Gemini CLI, Technical Prompt Engineering

Package Managers and Utilities

Chocolatey, WinGet, Scoop, Inno Setup, npm, Ventoy, Rufus, Package Management Automation

Additional Technologies

System Administration, Windows Installation and Recovery, UEFI/GPT Configuration, Windows Optimization, Developer Environment Setup, Single-file Browser Tools, File Extension Downloads, Client-side Caching

KEY PROJECTS AND TECHNICAL CAPABILITIES

Flutter Inventory Management System

Designed and built a multi-role inventory management system using Flutter and Supabase with Owner, Manager, and Employee role separation. Implemented comprehensive features including product management, category management, supplier management, customer management, purchase tracking, inventory tracking, POS/Sales functionality, and reporting dashboards. Developed offline synchronization architecture and PostgreSQL database schema with SQL migrations. Fixed race conditions in stock updates and implemented reporting with charts and data exports.

Cash Collection and POS Web Application

Developed a touch-friendly cash collection interface as a full-stack web application with Owner and Cashier workflows. Created responsive dashboard and transaction history features with customized receipt layout. Implemented cashier management capabilities and planned barcode scanner and receipt printer compatibility.

Pakeeza Pay - Payment Confirmation Web Application

Built a dual-QR payment confirmation system for bakery operations using HTML/CSS/JS frontend, Node.js/Express backend, and SQLite database. Implemented session-based authentication with bcrypt and express-session. Designed sequential QR display for student payment flow and created administrative dashboard with GoDaddy Core UI design theme (teal accent color).

Desktop Application Development

Converted Python scripts into production-ready desktop GUI applications using CustomTkinter and PyInstaller. Designed and structured database-backed applications featuring administrative controls and data import/export capabilities. Developed Windows PowerShell GUI utilities and Qt/C++ desktop applications. Experience in creating resizable interfaces and standalone executable deployment.

Peer-to-Peer File Sharing Application

Planned and architected a Flutter application for Windows and Android platforms featuring local network device discovery and device-to-device file transfer without internet dependency. Designed desktop-style application architecture for cross-platform compatibility.

Windows Automation and Installer Tools

Designed and deployed the Tools Installer using Inno Setup for batch project deployment. Created PowerShell automation tools for software installation, updates, and package management using Chocolatey and Winget. Developed developer setup scripts and VPNGate SSTP auto-connection automation with health scoring and fallback logic.

Media Processing and Automation

Developed music organizer automation tools using Mutagen metadata handling. Created image conversion utilities and FFmpeg-based media workflows. Built download automation using yt-dlp and aria2. Implemented batch file renaming with OOP architecture demonstrating inheritance, polymorphism, and encapsulation.

WordPress and E-Commerce Solutions

Customized WooCommerce stores with modified checkout behavior and custom plugin features. Implemented SEO optimization using RankMath and performance optimization with LiteSpeed Cache. Performed theme customization and integrated custom HTML/CSS/JavaScript components. Completed cross-platform e-commerce data synchronization project for product and inventory tracking.

Browser Utilities and Web Tools

Created single-file browser tools for QR code detection, link conversion, and file caching. Developed interactive games and emergency vault applications. Implemented client-side caching and custom file extension downloads. Built command-line execution tools for batch cloning and scripting.

Workflow Automation with n8n

Developed multi-stage data synchronization pipelines and automated media downloader using n8n low-code platform. Completed cross-platform e-commerce synchronization for inventory and product tracking.

EDUCATION

Department of Computer Science, University of Engineering and Technology, Taxila, Pakistan

Relevant Coursework: Object-Oriented Programming (OOP), Data Structures, Algorithms, Database Management, Software Engineering

ADDITIONAL COMPETENCIES

- Full-stack application architecture and design patterns
- Scalable database schema design and SQL optimization
- Role-based access control (RBAC) and authentication systems
- Inventory and POS workflow design and implementation
- Report generation and data export (CSV, PDF, Excel)
- API integration and REST principles
- CLI tooling and command-line interface design

- Local-first application design principles
- Windows system administration and optimization
- Cross-platform software development and deployment
- Responsive UI/UX implementation
- Software debugging and performance troubleshooting
- Version control workflow management
- Microservices and distributed system concepts
- Technical documentation and code maintainability

TOOLS AND PLATFORMS

Version Control: Git, GitHub, GitLab

Development Environments: Visual Studio Code, Qt Creator, Android Studio, Visual Studio

Databases: SQLite, PostgreSQL, Supabase

Cloud Services: Google Cloud Platform (GCP), Netlify

Automation: n8n, PowerShell, Python, Bash

Build Systems: CMake, Ninja, Gradle

Testing and Debugging: Playwright, Puppeteer, Selenium, Android Debug Bridge (ADB)